

TRINA MUKHERJEE

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Education

- **Computer Science And Engineering (Graduation)** :- Jain University (2023-2027) , CGPA : – 9.0
- **Oxford Public School** :- Higher Secondary (2021-2023) , Percentage scored : - 87%
- **Loreto Convent** :- Matriculation (2020) , Percentage Scored :- 92.4%

Skills

- **Languages:** Python , Java (*Data Structures and Algorithms*), JavaScript
- **Frontend:** React.js, Next.js, HTML, CSS, Bootstrap
- **Backend:** Flask, FastAPI, Node.js
- **Databases:** MySQL, MongoDB, PostgreSQL
- **AI/ML:** Scikit-Learn, TensorFlow, Keras, XGBoost, PyTorch
- **Tools:** Git, Docker, Kafka, Redis

Experience

- **PERSONIFY (AI INTERN) — 1STOP.AI (OCT 2024- JAN 2025)** :- 1. Developed CNN-based models for pet classification and landmark detection to perform image-based predictions
2. Applied data preprocessing, augmentation, and feature extraction to improve model generalization and accuracy
3. Performed hyperparameter tuning and model evaluation using validation techniques to optimize performance
Tech Stack: Python, TensorFlow, Keras, OpenCV
- **CRCE HACKATHON - CHROME EXTENSION (SEPT 2025)** :- 1. Built CartHero, an AI-powered Chrome extension for eco-impact analysis and sustainable product recommendations
2. Integrated ML + real-time web scraping, achieving ~67% accuracy and ~73% emission reduction potential
3. Designed UI/UX and deployed the application for seamless end-to-end functionality
4. Collaborated in a 4-member team for development and hackathon presentation
Tech Stack: Python, Flask, Machine Learning, Web Scraping, Chrome Extension APIs
GitHub : - <https://github.com/TrinaMukherjee21/Green-Guardian-Cart-Extension->
(Live Demo :- <https://green-guardian-cart-extension.vercel.app/>)

Projects

- **ALPHASIGNAL – AI-POWERED TRADE SENTIMENT DASHBOARD** :- 1. Built a real-time trading intelligence platform that aggregates financial news, social media discussions, and market data to generate **BUY, SELL, and HOLD** signals for US and Indian stocks.
2. Developed a scalable data pipeline using FastAPI, Kafka, Redis, and PostgreSQL to process and stream live market data with low latency.
3. Implemented NLP-based sentiment analysis using **FinBERT (HuggingFace Transformers)** and **VADER** to derive confidence-based trading recommendations from financial text.
4. Created an interactive **React** dashboard with real-time sentiment tracking, market visualizations, and live signal updates using **Recharts, Zustand, and WebSockets**.
Tech Stack: React.js, FastAPI, Apache Kafka, PostgreSQL, Redis, WebSockets, FinBERT, VADER, PyTorch, Docker
GitHub : - <https://github.com/TrinaMukherjee21/AlphaSignal->
(Live Demo :- <https://alpha-signal-two.vercel.app/>)
- **STYLE GENIE : AI-POWERED FASHION ASSISTANT** :- 1. Developed an AI-powered fashion platform featuring a scalable product catalog, curated collections, and personalized discovery experiences.
2. Implemented Virtual Try-On and color analysis modules using AI to enable real-time outfit visualization and personalized

style recommendations.

2. Built an intelligent styling assistant with personalized user features, including context-aware guidance, lookbook, and wishlist for enhanced user engagement.

Tech Stack: Python, Flask, Node.js, React, Next.js, Tailwind CSS, JavaScript, Gradio, PIL, OpenAI API, Google Gemini API, Hugging face, Scikit-learn, FAISS, Serper API, JSON.

GitHub : - <https://github.com/TrinaMukherjee21/Style-Genie->

(Live Demo :- <https://stylegenie-frontend.onrender.com>)

- **BANK FRAUD DETECTION SYSTEM** :- 1. Built a fraud detection system using XGBoost + SMOTE-ENN, achieving ~98.9% ROC- AUC and 98.6% accuracy with optimized thresholds
2. Developed a Flask-based dashboard with RBAC and JWT authentication for secure, real-time fraud monitoring using Chart.js
3. Implemented a low-latency REST API (~85ms) with audit logging and risk analysis to ensure data integrity and compliance .

Tech Stack: Python, Flask, XGBoost, CatBoost, Scikit-learn, Imbalanced-learn, PyJWT, Flask-Login, SQLite/MySQL, HTML, CSS, Jinja2, Chart.js

GitHub : - <https://github.com/TrinaMukherjee21/Bank-Fraud-Detection->

(Live Demo :- <https://bank-fraud-detection-1-cfcb.onrender.com/login?next=%2F>)

Additional Projects

- **OMNISALES : CUSTOMER RELATIONSHIP MANAGEMENT (CRM) SYSTEM** :- 1. Built an interactive CRM analytics dashboard to analyze retail transaction data and deliver actionable business insights through real-time reporting
2. Applied K-Means clustering and Decision Tree models to enable customer segmentation and purchase prediction, supported by dynamic visualizations of revenue and sales metrics.

Tech Stack: Python, Streamlit, Pandas, NumPy, Scikit-learn, Matplotlib

GitHub : - <https://github.com/TrinaMukherjee21/OmniSales>

(Live Demo :- <https://omnisales-fsc5s8bi6wn35gyf8asijx.streamlit.app/>)